

Abstract

Road crash records guide decisions about where enforcement and engineering money goes, so their quality needs checking before any model is fitted. This study examined a public Kaggle file of Bangladesh road accidents ($N = 47,676$ after duplicate removal; 31 variables) using exploratory analysis, chi-square tests of association, and supervised classification. The outcome was whether an accident recorded at least one death, which applied to 66.8% of records. Recorded accidents rose by 98.8% between 2007 and 2021, while the proportion of fatal accidents stayed between 65.8% and 67.5%. None of 18 road, light, weather, and calendar variables was significantly associated with fatality (largest $V = .026$), and a tuned classifier evaluated on 9,536 held-out accidents reached an area under the ROC curve of .504, indistinguishable from chance. Data checks pointed to a likely cause. Death counts were split almost evenly across 0, 1, and 2–4 deaths, category sizes were nearly identical, the date column conflicted with the year column, and most times of day converted to a single hour. These features are consistent with synthetic or heavily edited records. The file is unsuitable for risk-factor inference. The paper offers an audit procedure for crash data and a minimum list of fields for future collection.

Keywords: road safety, exploratory data analysis, data quality, chi-square test, Cramér's V , classification

Auditing a Public Road Accident Dataset from Bangladesh: Exploratory Analysis, Association Testing, and Fatality Classification

Road traffic crashes kill about 1.19 million people each year, according to the World Health Organization (WHO, 2023). The WHO South-East Asia Region, which includes Bangladesh, accounted for 28% of those deaths. Decisions about where to place police, which junctions to re-design, and which campaigns to run depend on crash records, and conclusions drawn from those records are only as good as the records.

Public datasets make crash analysis open to students, journalists, and researchers who have no access to police files. They also arrive without the information needed to judge them. A spreadsheet rarely says who recorded each crash, how fatalities were counted, or whether values were edited after collection. The dataset examined here, titled *Bangladesh Road Accident Dataset 2007-2024* (mdnahidurrahmankh, n.d.), is an example. It has 47,680 rows and 31 columns, and the file loaded for this study was named `Modified_Accident_Dataset_final.xlsx`.

The study began as an attempt to answer practical safety questions and turned into an audit of the data. Four research questions guided the work:

1. How do the volume and fatality of recorded accidents vary across years and calendar periods?
2. Is fatality associated with any recorded road, light, weather, or calendar variable?
3. Can a supervised classifier predict whether an accident is fatal from those variables?
4. Do internal consistency checks support using the file for the first three questions?

The answers were negative for the second and third questions and pointed to problems for the fourth. The sections below describe the method, report each result, and discuss what the pattern means for people who use this file or collect similar data.

Method

Data

The data came from the public Kaggle dataset cited above. The spreadsheet contained 47,680 rows and 31 variables covering accident identifiers, date and time, location text, driver age, vehicle information, death and injury counts, light and weather conditions, and several road descriptors. Table 1 groups the variables by topic. Weather appears in two columns (weather_x, weather_y) and so does lighting (light_condition, lighting). The suffixes _x and _y are the ones pandas adds when two tables with a shared column name are merged, which suggests the file combines two sources.

Table 1

Variables in the Accident Dataset Grouped by Topic

Topic	Variables
Identification	accident_serial_no, id
Time	date, time, year, month
Place	location, longitude_and_latitude, place_characteristic
People and vehicles	driver_age, vehicle_info, vehicle_count, victim_category
Outcome	death_count, death_info, injury_count, injury_info, accident_intensity
Narrative	accident_narrative
Road	road_surface_type, road_nature, road_classification, type_of_road_connection_point, road_surface_condition, junction, traffic_control, traffic_condition
Light and weather	light_condition, lighting, weather_x, weather_y

Note. Variable names are shown after conversion to lower case with underscores. The spelling traffic_condition is as it appears in the file.

Preprocessing

Column names were converted to lower case and underscores, whitespace was trimmed from text values, and exact duplicate rows were removed. Four duplicates were found, leaving 47,676 accidents. No variable had missing values.

The date column mixes ISO timestamps (for example, 2022-06-05 00:00:00) with day-month-year strings (for example, 13/6/2022). The conversion routine accepted a date column only

when more than half of its values could be parsed, and this column did not meet that criterion. Weekday and season were therefore not derived, and the file’s own year and month columns were used instead. Times were converted to an hour of the day from strings such as 06.35 am and 08:20 AM.

Outcome Definition

An accident was classified as *fatal* when `death_count` was greater than zero. Accidents were also placed in severity classes by number of deaths (0, 1, 2–4, and 5 or more). Total casualties were the sum of deaths and injuries.

Statistical Analysis

Each categorical variable with two to 30 categories and no outcome information was cross-tabulated with the fatal indicator, and a Pearson chi-square test of independence was run at $\alpha = .05$. Year and hour were treated as categorical for this purpose. Because chi-square tests on large samples detect very small differences, the strength of each association was measured with Cramér’s V (Cramér, 1946). Every table had two columns (fatal and not fatal), so V equals Cohen’s (1988) w , for which values of .10, .30, and .50 mark small, medium, and large effects. Tests were run with SciPy (Virtanen et al., 2020).

Predictive Modeling

A classifier was trained to predict the fatal indicator from the calendar variables and the road, light, and weather variables. Death counts, injury counts, casualty totals, and other outcome-related columns were excluded to prevent leakage. The data were split 80/20 with stratification on the outcome, leaving 9,536 accidents for testing. Three model families from scikit-learn (Pedregosa et al., 2011) were compared with 5-fold stratified cross-validation on the area under the receiver operating characteristic curve (AUC; Fawcett, 2006) and macro-averaged F1: logistic regression, a random forest, and histogram-based gradient boosting. Class weights were balanced because fatal accidents outnumbered the rest about two to one. The best family was tuned with a randomized search of 15 parameter draws, and the final model was scored once on the held-out set. The random seed was 42. The cross-validation scores of the individual families were not saved, so

the winning family cannot be named.

Software

Data handling used pandas (McKinney, 2010), and figures were drawn with matplotlib (Hunter, 2007) and seaborn (Waskom, 2021).

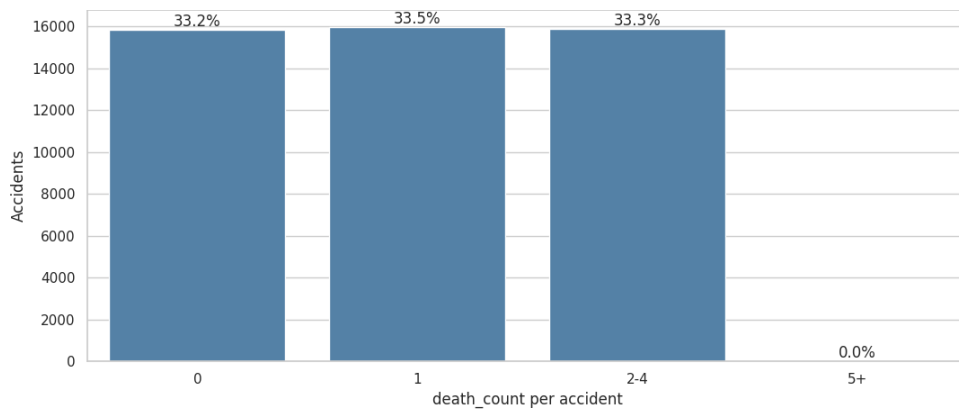
Results

Descriptive Overview

Accident severity was spread almost evenly across the first three classes (see Figure 1). Accidents with no deaths made up 33.2% of the file, accidents with one death 33.5%, and accidents with two to four deaths 33.3%. The class with five or more deaths rounded to 0.0%. Overall, 66.8% of accidents were fatal.

Figure 1

Accidents by Number of Deaths



Note. Percentages above the bars show each class's share of 47,676 accidents. The first three classes differ by less than one percentage point.

Two features of this distribution are unusual. Crash records commonly show fewer events as the number of deaths per crash rises, and here the first three classes are almost the same size. The file also contains no accident with five or more deaths in nearly 48,000 records.

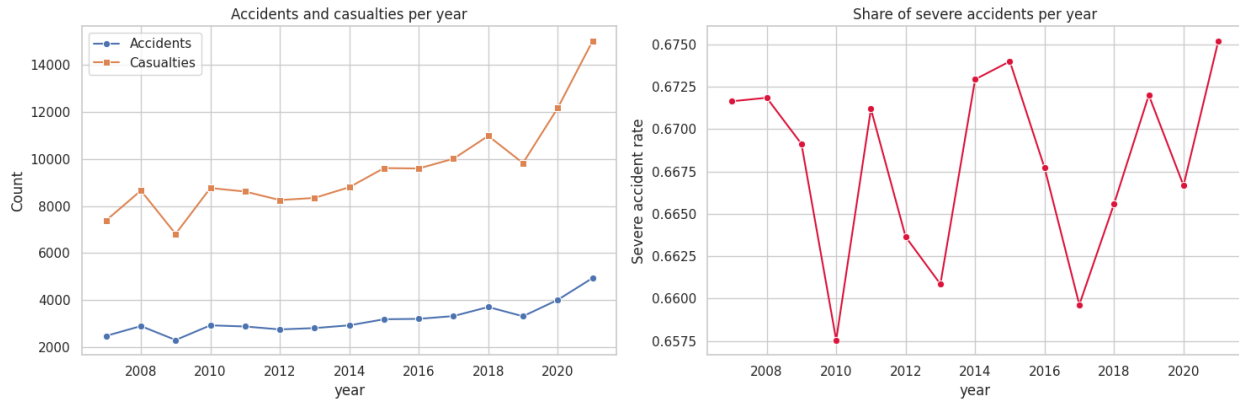
Trend Across Years

The year variable ran from 2007 to 2021, not to 2024 as the dataset title states. Recorded accidents rose from roughly 2,500 in 2007 to roughly 4,900 in 2021, an increase of 98.8%, and a linear fit added 123.6 accidents per year. Total casualties grew from roughly 7,400 to 15,000 over

the same span and peaked in 2021 (see Figure 2). The fatal proportion barely changed. Across the 15 years it stayed between 65.8% and 67.5%, so the trend lies in volume only.

Figure 2

Recorded Accidents, Casualties, and Fatal Proportion by Year



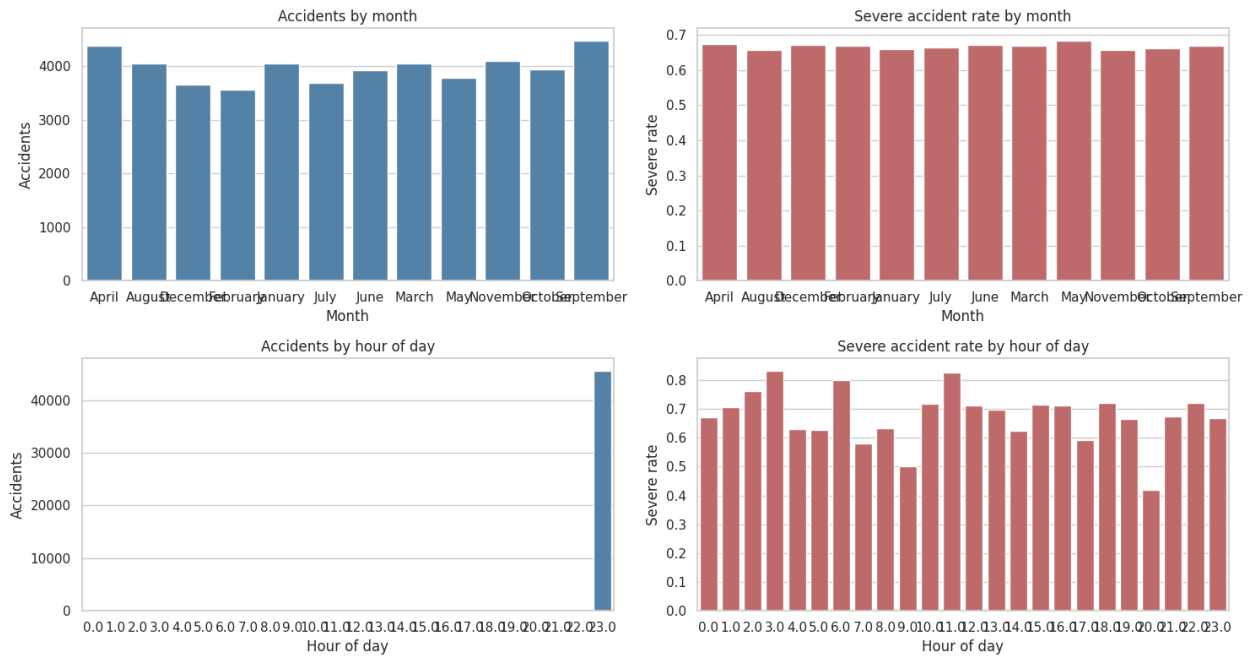
Note. Left panel: annual counts of accidents and casualties (deaths plus injuries). Right panel: proportion of accidents with at least one death, labeled “severe accident rate” in the axis. The vertical axis of the right panel spans only about 1.7 percentage points.

The counts fall in 2009 and 2019 before rising sharply. The 2020 count sits above 2019, which may seem odd given travel restrictions that year in many countries, although the file offers no way to check whether that reflects real traffic, changes in reporting, or the way the file was assembled. The doubling of records should therefore not be read as a doubling of crashes.

Calendar Patterns

Monthly counts ranged from about 3,550 accidents in February to about 4,480 in September, so September had the most accidents. The fatal proportion was highest in May (68.5%), and the other months lay close to 66%–68% (see Figure 3).

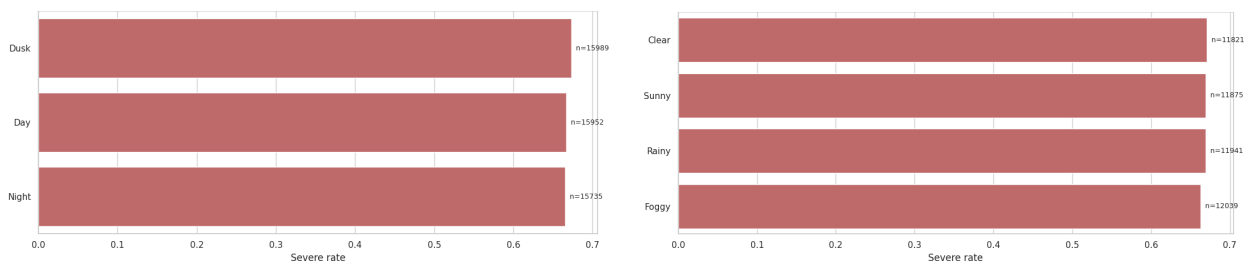
The hour-of-day results could not be used. Nearly every accident fell in hour 23, with only a few hundred records spread across the other 23 hours. Traffic does not behave this way, so either the time strings were mostly stored in a form that converts to 23:xx or the field holds a placeholder for most rows. The derived day-part comparison is affected in the same way. The notebook output reported a fatal proportion of 69.0% for morning and 66.8% for evening accidents, but the evening group held almost every record, and that comparison carries no information.

Figure 3*Accidents and Fatal Proportion by Month and by Hour of Day*

Note. Months appear in alphabetical order because the file stores them as text. The lower left panel shows almost all accidents in hour 23. The right panels are labeled “severe rate” in the axes.

Road, Light, and Weather Conditions

Fatal proportions barely differed across environmental conditions (see Figure 4). Dusk, day, and night accidents were fatal between 66.5% and 67.3% of the time. For clear, sunny, rainy, and foggy weather the range was 66.2% to 67.0%. Night and fog did not show higher fatal proportions than daylight and clear skies.

Figure 4*Fatal Proportion by Light Condition and by Weather*

Note. Left: light condition. Right: weather (weather_x). Labels show the number of accidents in each category. Horizontal axes are labeled “severe rate.”

Category sizes were also strikingly even. The three light categories held 15,735 to 15,989

accidents each, and the four weather categories held 11,821 to 12,039 each. Table 2 shows the opposite pattern for two road variables, where a single value covers nearly every row. The only visible gap was for two-lane roads ($n = 1,281$), where the fatal proportion was about 69%, compared with about 67% on roads recorded as lanes. The association tests below show that this gap is within chance variation.

Table 2

Accident Counts by Road Surface Type and Road Nature

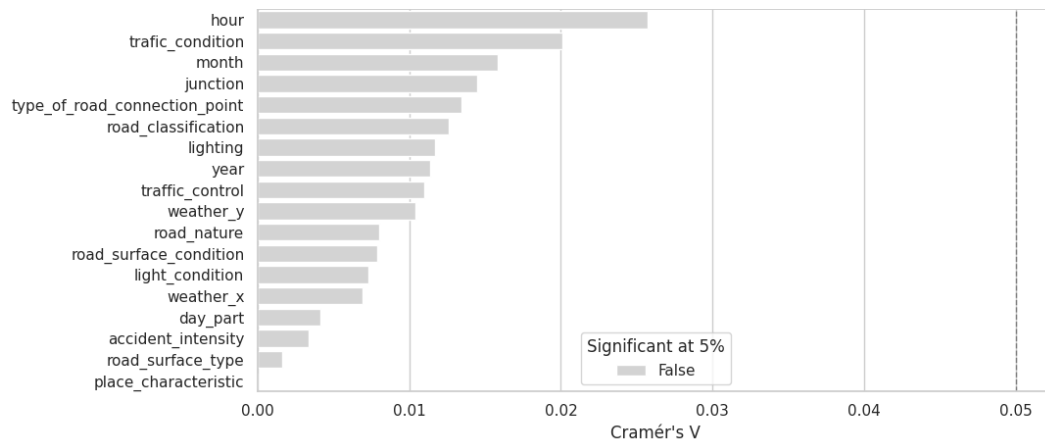
Variable	Value	<i>n</i>
road_surface_type	Paved	47,675
	Brick Road	1
road_nature	Lane	46,389
	Two-Lane	1,281
	Two-lane	6

Note. “Two-Lane” and “Two-lane” are stored as separate values in the file.

Association With Fatality

Eighteen variables entered the association tests. None was significant at the .05 level (see Figure 5 and Table A1 in Appendix A). The closest were road classification, $p = .056$, and lighting, $p = .087$. The largest effect was for hour of day, $V = .026$, followed by traffic condition, $V = .020$, and month, $V = .016$. All values fall below .10, the lower bound of Cohen’s (1988) small effect. The outcome-like variable `accident_intensity` had $V = .003$, $p = .907$, so it was practically unrelated to the death count it might be expected to track.

With 47,676 records, the tests had ample power to detect small differences, and none appeared. At a .05 threshold, 18 independent tests would produce about one false positive by chance alone, and the observed count of zero is in line with variables that are unrelated to the outcome. The hour result has an additional caveat, since it comes from the column whose values collapsed into a single hour.

Figure 5*Strength of Association Between Each Variable and Fatal Outcome*

Note. Bars show Cramér's V. The dashed line marks .05. No variable was significant at $\alpha = .05$, so all bars are shown in the same light shade.

Classification of Fatal Accidents

On the 9,536 held-out accidents, the tuned model reached an AUC of .504. Table 3 gives the full classification report. An AUC of .50 means the model ranks a fatal accident above a non-fatal one half of the time, the same as guessing. Precision matches the class base rates. Fatal accidents made up 66.8% of the test set and precision for that class was .67, while non-fatal accidents made up 33.2% and precision there was .33. Accuracy was .49, below the .668 that a rule predicting every accident as fatal would reach, because balanced class weights spread the predictions across both classes.

Table 3*Classification Report on the Held-Out Test Set*

Class	Precision	Recall	F1	Support
Not fatal	.33	.53	.41	3,164
Fatal	.67	.47	.55	6,372
Accuracy			.49	9,536
Macro average	.50	.50	.48	9,536
Weighted average	.56	.49	.50	9,536

Note. AUC = .504. The model was tuned on the training portion only.

The result does not reflect a weak model. Tuning cannot recover signal that the inputs do

not hold, and the association tests had already shown that the inputs hold none.

Data Quality Audit

Flat results could mean that fatality really is independent of the recorded conditions. For light and road conditions this seemed unlikely, so the file itself was examined for a second explanation. Table 4 lists the indicators found.

Table 4
Indicators of Data Quality Problems

Indicator	Evidence
Even outcome classes	Deaths of 0, 1, and 2–4 held 33.2%, 33.5%, and 33.3% of accidents; the 5+ class was empty after rounding.
Even category sizes	Light condition: 15,735 to 15,989 per category. Weather: 11,821 to 12,039 per category.
Conflicting dates	The first row has the date 2022-06-05, the year 2007, and the month September.
Collapsed hours	Nearly every time converted to hour 23.
Repeated descriptions	In the first five rows, driver age was 26 each time, and the death description named the same 50-year-old man in all five, including rows with a death count of zero.
Fields and narrative disagree	Row 1 gives a time of 06.35 am and a car–pedestrian collision, while its narrative describes an accident around 11:45 AM involving a loaded cargo truck.
Unrelated outcome variable	accident_intensity was unrelated to fatality ($V = .003$).
Incomplete coverage	The data ended in 2021, not 2024.
Inconsistent labels	“Two-Lane” and “Two-lane” were stored as separate values.
Modified source	The file name contains the word “Modified.”

Note. Rows refer to the cleaned file of 47,676 accidents.

No single indicator is decisive. Together they point the same way: many values seem to have been assigned independently of one another, which would explain why no variable predicts any other. The even category sizes are the clearest sign. Values drawn at random from a short list of categories produce near-equal counts, as seen here, whereas real traffic piles up in some conditions and thins out in others. The word “Modified” in the file name suggests an edited original, and the _x and _y columns suggest a merge. How the file was produced could not be verified.

Discussion

The study found one robust pattern and several null results. Recorded accidents roughly doubled between 2007 and 2021, but the fatal proportion stayed flat, no road, light, weather, or calendar variable was associated with fatality, and a tuned classifier performed at chance. Inspection of the file showed internal contradictions and implausibly even distributions. The most economical reading is that the file does not preserve the relationships found in real crash data, so its null results say nothing about Bangladeshi roads.

Implications

Anyone planning to use this dataset should treat it as practice material for analytic methods and not as evidence about road risk. The doubling of records and the lack of any condition effect describe the file. They do not describe traffic.

For analysts working with any crash dataset, the audit steps used here cost little and would have flagged the problems at the start. The outcome distribution should be plotted first, because real severity counts fall as deaths per crash rise. Date-like columns should be compared with each other, since year, month, and a full date ought to agree. Rows per category should be counted, because near-identical sizes across unrelated categories suggest random assignment. Each variable should be tested against the outcome with an effect size as well as a p value. A sample of full rows, narrative included, should be read to see whether the fields agree. Finally, yearly totals should be compared with an official source before any conclusion is drawn.

For people who collect crash data, a file that could support safety decisions would record a verified date and a 24-hour time, district and division, checked coordinates, vehicle types and road user types, road class and speed limit, a consistent injury scale, the outcome after a fixed period such as 30 days, and the reporting agency. With those fields, location risk tiers and time-window risk classes become possible. In the present study a planned district risk classification produced no output because the file has no district or division variable, only free-text locations.

Limitations

The study rests on one file whose origin could not be verified, so every finding describes the file. The outcome was a binary fatal indicator, and injury severity and number of deaths were not modeled. Free-text columns (narrative, location, and coordinate strings) were not mined. Parsing coordinates into districts is an obvious extension but is worth doing only on data that pass the audit. Hour and day-part results are unreliable because of the time collapse, and weekday and season results do not exist because the date column failed to parse. The classifier received a modest tuning budget of 15 draws, although with an AUC at chance a larger budget would be unlikely to change the conclusion. The classification results come from an earlier version of the analysis notebook, and the cross-validation scores of the individual model families were not retained. Association tests describe relationships between variables and cannot show cause, even in a clean file.

Future Research

The next step is to obtain a verified crash dataset, for example from police records, the Bangladesh Road Transport Authority, or university research units, and to repeat the analysis. Comparing the yearly totals in this file with official statistics would show how far the records diverge from reported counts. On clean data, the district-level risk classification, time-window classification, and severity models described here could be run without change.

Conclusion

The study set out to identify risk factors for fatal road accidents in Bangladesh and ended by showing that the available file cannot support such an analysis. A classifier with an impressive score on these data would have looked like a finding and would have been wrong. Checking a dataset before modeling it is the more valuable result.

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Appendix A

Chi-Square Tests of Each Variable Against Fatal Outcome

Table A1

Association of Each Variable With Fatal Outcome, Ordered by Effect Size

Variable	Categories	V	p	Significant
hour	24	.0257	.117	No
traffic_condition	25	.0201	.741	No
month	12	.0159	.364	No
junction	9	.0145	.268	No
type_of_road_connection_point	7	.0134	.198	No
road_classification	4	.0126	.056	No
lighting	4	.0117	.087	No
year	15	.0114	.961	No
traffic_control	6	.0110	.330	No
weather_y	4	.0105	.157	No
road_nature	3	.0080	.217	No
road_surface_condition	4	.0079	.396	No
light_condition	3	.0073	.279	No
weather_x	4	.0070	.511	No
day_part	4	.0041	.848	No
accident_intensity	4	.0034	.907	No
road_surface_type	2	.0016	.721	No
place_characteristic	2	.0000	1.000	No

Note. V = Cramér's V. Significance refers to $\alpha = .05$. The analysis notebook listed month twice because the column was picked up both as a text variable and as an added calendar field; it is shown once here.